# Control work 3, and work on the mistakes

Approximation and modelling in one paper — the two topics where a missing unit costs a mark.

LESSON 35–36 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 3 | P1 · CHAPTER 9 REVIEW

**LESSON CLOCK**

3 | 40' | 10' | 25' | 12'

| | |
|---|---|
| Instructions | 3 min |
| The paper | 40 min |
| Self-mark | 10 min |
| Rewrite | 25 min |
| Units drill | 12 min |

**LEARNING OBJECTIVES**

- Apply linear approximation and connected rates under time.
- Give every answer with its units and in context.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

**TEXTBOOK**

| | |
|---|---|
| National | pp. 169–172 |
| Cambridge | Pure Mathematics 1, pp. 191–194 |
| Workbook | Control paper C |

## 01 Explanation

### The paper — 25 marks, 40 minutes

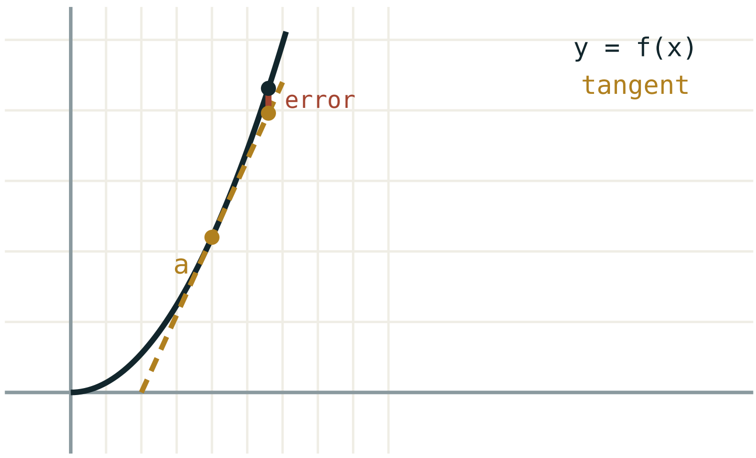
| Q | TASK | MARKS | FROM |
|---|---|---|---|
| 1 | Estimate $\sqrt{37}$ and $2.02^4$ by linear approximation | 5 | L28–30 |
| 2 | A cube's side is $12 \pm 0.03$ cm. Find the volume and its percentage error | 4 | L28–30 |
| 3 | $s = t^3 - 9t^2 + 24t$: find when the body is at rest and the distance in the first 5 s | 6 | L31–34 |
| 4 | A balloon inflates at $40$ cm³/s. Find $\frac{\mathrm{dr}}{\mathrm{dt}}$ when $r = 4$ | 5 | L31–34 |
| 5 | $C(q) = 0.04q^2 + 15q + 600$: marginal and average cost at $q = 50$ | 5 | L31–34 |

UNITS ARE MARKED

Q3, Q4 and Q5 each carry one mark for the units alone. A numerically perfect answer with no units loses three of the twenty-five.

## The three errors this topic produces

| ERROR | LOOKS LIKE | KIND |
|---|---|---|
| units omitted | “the answer is 0.4” | careless |
| two variables kept | differentiating $V = \frac{1}{3}\pi r^2 h$ in $t$ without eliminating $r$ | method |
| distance read as displacement | ignoring the change of direction | method |
| wrong $a$ chosen | estimating $\sqrt{37}$ from $a = 25$ | knowledge |




*Question 1 in one picture — and the reason the choice of a matters.*

## The units drill

Ten rates on the board. For each, the class calls out only the **units**, in three seconds:

| RATE | UNITS |
|---|---|
| $\frac{ds}{dt}$, $s$ in km, $t$ in hours | *km/h* |
| $\frac{dV}{dt}$, $V$ in litres | *litres per second* |
| $\frac{dC}{dq}$, cost in so‘m | so‘m per item |
| $\frac{dA}{dr}$, $A$ in cm² | *cm* |
| $\frac{dP}{dt}$, $P$ a population | people per year |

ROW 4 IS THE INTERESTING ONE

$\frac{dA}{dr} = 2\pi r$ has units cm²/cm = cm. A rate need not be “per second”: it is whatever the two quantities make it.

## 02 Worked examples

EXAMPLE 1 **Model answer, Q1: estimate $\sqrt{37}$.**

1. $a = 36, h = 1$

   Nearest convenient square.

2. $f(36) = 6, f'(36) = \frac{1}{12}$

3. $6 + 1 \times \frac{1}{12} \approx 6.083$

   True $6.0828$.

ANSWER

$\approx 6.083$

EXAMPLE 2 **Model answer, Q4: balloon at $40$ cm³/s, find $\frac{dr}{dt}$ at $r = 4$.**

1. $V = \frac{4}{3}\pi r^3$

   $\frac{dV}{dr} = 4\pi r^2 = 64\pi$

2. $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

3. $40 = 64\pi \cdot \frac{dr}{dt}$

4. $\frac{dr}{dt} = \frac{40}{64\pi} \approx 0.199$

ANSWER

$\approx 0.20$ cm/s

EXAMPLE 3 **A learner wrote “the top of the ladder falls at 0.3”. What is missing?**

1. The number is right.

2. No units, and no direction.

   A careless error.

3. Correct: $0.3$ m/s downwards.

ANSWER

The units and the direction

## 03 Interactive model

Run the units drill before the rewrite, not after.

## Units and estimates

Best $a$ for estimating $\sqrt{37}$:

- *25*
- *36*
- *40*
- *49*

$\sqrt{37} \approx$

- *6.05*
- *6.083*
- *6.2*
- *6.5*

Units of $\frac{dA}{dr}$ for $A$ in cm²:

- $cm^2$
- $cm$
- $cm^3$
- $cm^2/s$

Balloon at 40 cm³/s, $r = 4$: $\frac{dr}{dt} \approx$

- $0.20$
- $0.40$
- $2.0$
- $6.4$

$s = t^3 - 9t^2 + 24t$: rest at

- $t = 2, 4$
- $t = 1, 3$
- $t = 3, 8$
- $t = 0, 6$

A 0.25% error in a cube's side gives what error in the volume?

- 0.25%
- 0.5%
- 0.75%
- 3%

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Control work | Nazorat ishi | Контрольная работа |
| Linear approximation | Chiziqli yaqinlashish | Линейное приближение |
| Connected rates | Bog‘liq tezliklar | Связанные скорости |
| Units | O‘lchov birliklari | Единицы измерения |
| Interpretation | Talqin | Интерпретация |
| Careless error | E’tiborsizlik xatosi | Ошибка по невнимательности |
| Method error | Usul xatosi | Ошибка в методе |
| Knowledge gap | Bilim bo‘shlig‘i | Пробел в знаниях |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

## Check yourself

A rate’s units come from:

- the numerator only
- both quantities
- the context
- nothing

Before differentiating a connected-rate formula you must:

- square it
- reduce it to one variable
- integrate
- nothing

Distance differs from displacement when:

- never
- the velocity changes sign
- the acceleration is zero
- always

Choosing a poor $a$ in an estimate is:

- careless
- a method error
- a knowledge gap
- fine

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up 7 PROBLEMS

1. Estimate $\sqrt{37}$

   ANSWER $\approx 6.083$

2. Estimate $2.02^4$

   ANSWER $\approx 16.64$

3. Units of $\dfrac{ds}{dt}$ for km and hours

   ANSWER $km/h$

4. $s = t^2 - 4t$; when at rest?

   ANSWER $t = 2$

5. $C(q) = 5q + 90$; marginal cost

   ANSWER $5$

6. $V = \dfrac{4}{3}\pi r^3$; $\dfrac{dV}{dr}$ at $r = 4$

   ANSWER $64\pi$

7. A 1% error in the side gives what error in the area?

   ANSWER $\approx 2\%$

## Medium · the standard question

7 PROBLEMS

1. Cube side $12 \pm 0.03$; percentage error in the volume

   ANSWER $0.75\%$

2. $s = t^3 - 9t^2 + 24t$; rest at

   ANSWER $t = 2, 4$

3. Same; distance in the first 5 s

   ANSWER $28$ m

4. Balloon at 40 cm³/s; $\dfrac{\mathrm{dr}}{\mathrm{dt}}$ at $r = 4$

   ANSWER $\approx 0.20$ cm/s

5. $C(q) = 0.04q^2 + 15q + 600$; marginal cost at $q = 50$

   ANSWER $19$

6. Same; average cost at $q = 50$

   ANSWER $\approx 32$ so‘m

7. Estimate $\dfrac{1}{5.02}$

   ANSWER $\approx 0.1992$

## Hard · stretch and reason

7 PROBLEMS

1. Estimate $\sqrt[3]{27.5}$

   ANSWER $\approx 3.0185$

2. A cone of height 12, radius 6, fills at 4 m³/min. $\frac{dh}{dt}$ at $h = 6$

   ANSWER $\frac{4}{9\pi} \approx 0.141$ m/min

3. $s = t^4 - 4t^3$; find every instant of rest

   ANSWER $t = 0, 3$

4. A square's area grows at 8 cm²/s. Find $\frac{da}{dt}$ at $a = 4$

   ANSWER $1$ cm/s

5. A ladder 10 m long; the top falls at 0.6 m/s. Find the foot's speed when the top is 6 m up

   ANSWER $0.45$ m/s

6. $R(q) = 80q - 0.4q^2$, $C(q) = 20q + 500$. Maximise the profit

   ANSWER $q = 75$, profit $1750$

7. Show that for $y = x^n$ the percentage error multiplies by $n$

   ANSWER $\frac{\Delta y}{y} = n\frac{\Delta x}{x}$

## 07 Homework

### Homework — 4 tasks

Task 1 is the rewrite. Units on every line.

1. Rewrite in full every question that lost a mark, with units on every answer.
2. Five problems from the section your knowledge column was heaviest in.
3. Estimate $\sqrt{50}$ and $3.01^5$.

4. A cone of height 20 and top radius 10 fills at $5$ cm³/s. Find $\dfrac{dh}{dt}$ when $h = 8$.